

## Recursion Problem List

Write a program that

1. Prints the numbers from 1 to n.

```
#include <iostream>
using namespace std;
void func(int i, int n)
{
    if(i>n) return;
    else
    {
        printf("%d ", i);
        func(i+1, n);
    }
}

void printN(int n)
{
    if(n<=0)
        return;
    else
    {
        printN(n-1);
        cout << n << " ";
    }
}

int main()
{
    printN(5);
}
```

2. Calculate the sum of numbers from 1 to n.

```
#include <iostream>
using namespace std;
int sum(int n)
{
    if(n==0) return 0;
    else
    {
        int _friend = sum(n-1);
        int result = n + _friend;
        return result;
    }
}

int main()
{
    int result = sum(10);
}
```

```
    cout << result << endl;
}
```

3. Calculate the factorial of n.

```
#include <iostream>
using namespace std;
int fact(int n)
{
    if(n==0) return 1;
    else
    {
        int _friend = fact(n-1);
        int result = n * _friend;
        return result;
    }
}
int main()
{
    int result = fact(5);
    cout << result << endl;
}
```

4. Calculate the sum of digits of a given number n.

```
#include <iostream>
using namespace std;

int sumOfDigit(int n)
{
    if(n==0) return 0;
    else
    {
        int lastDigit = n % 10;
        n = n / 10;
        int _friend = sumOfDigit(n);
        return _friend + lastDigit;
    }
}

int main()
{
    int result = sumOfDigit(2489);
    cout << result << endl;
}
```

5. Count the number of digits of a given number n.

```

#include <iostream>
using namespace std;
int countDigit(int n)
{
    if(n==0) return 0;
    else
    {
        n = n / 10;
        int _friend = countDigit(n);
        return _friend + 1;
    }
}
int main()
{
    int result = countDigit(2489);
    cout << result << endl;
}

```

6. Calculate the  $n^{\text{th}}$  term of a Fibonacci series.

```

#include <iostream>
using namespace std;
int fib(int n)
{
    if(n==1) return 0;
    else if(n==2) return 1;
    else
    {
        int prevTerm = fib(n-1);
        int prevPrevTerm = fib(n-2);
        return prevTerm + prevPrevTerm;
    }
}
int main()
{
    cout << fib(8) << endl;
}

```

7. Calculate a to the power b.

```

#include <iostream>
using namespace std;
int pow(int a, int b)
{
    if(b==0) return 1;
    else
    {
        int _friend = pow(a, b-1);
        return a * _friend;
    }
}

```

```

    }
}
int main()
{
    int a, b; cin >> a >> b;
    int result = pow(a, abs(b));

    if(b<0)
        cout << 1.0/result << endl;
    else
        cout << result << endl;
}

```

8. Print the array elements.

```

#include <iostream>
using namespace std;
void printArray(int arr[], int n, int idx)
{
    if(idx==n)
        return;
    else
    {
        cout << arr[idx] << " ";
        printArray(arr, n, idx+1);
    }
}
int main()
{
    int arr[10] = {1, 4, 7, 2, 5, 8, 3, 6, 9, 0};
    int n = 10;

    printArray(arr, n, 0);
}

```

9. Find the largest element of a given array.
10. Find the smallest element of a given array.
11. Find the largest and smallest element of a given array.
12. Check whether a given string is palindrome or not.
13. Remove white spaces from a string & convert upper cases to lower.

```

#include <iostream>
using namespace std;
string removeWSCaseConversion(int idx, string str)
{
    if(idx==str.length())
        return "";
    else

```

```

{
    char currentCharacter = str[idx];

    /// delegating the work
    string _friend = removeWSCaseConversion(idx+1, str);

    /// doing my own work here
    if(currentCharacter>='A' && currentCharacter<='Z')
    {
        currentCharacter-='A';
        currentCharacter+='a';
    }

    if(currentCharacter==' ')
        return _friend;
    else
        return currentCharacter + _friend;
}

}

int main()
{
    string res = removeWSCaseConversion(0, "Abeer Hassan");
    cout << res << endl;
}

```